

USE-CASE FLOW SPECIFICATION

Use Case: Record Grant Expense

1. Use Case Name

Record Grant Expense

2. Primary Actor

Faculty Researcher

3. Goal

To allow the Faculty Researcher to record an equipment procurement expense against a research grant and update the remaining grant balance.

4. Preconditions

1. The Faculty Researcher is logged into the system.
2. A valid research grant exists for the Faculty Researcher.
3. The Faculty Researcher has permission to record expenses for the grant.

5. Main Success Scenario

1. The Faculty Researcher selects a research grant.
2. The system displays the grant details, including the allocated amount and remaining balance.
3. The Faculty Researcher selects Record Grant Expense.
4. The system displays the expense entry form.
5. The Faculty Researcher enters the equipment procurement expense details and amount.
6. The system validates the entered expense details.
7. The system checks whether the expense is within the available grant balance.
8. The system records the expense.
9. The system calculates the updated remaining grant balance.
10. The system displays the updated remaining grant balance to the Faculty Researcher.
11. The use case ends successfully.

6. Alternate Flow

A1. Expense Exceeds Available Grant Balance

1. At Step 7, the system determines that the expense exceeds the available grant balance.
2. The system rejects the expense.
3. The system displays a message informing the Faculty Researcher that the expense cannot be recorded because it exceeds the available grant balance.
4. The expense is not recorded.
5. The existing grant balance remains unchanged.
6. The use case ends.

7. Postconditions

Success: The expense is recorded successfully, and the remaining grant balance is updated.

Failure: The expense is not recorded, and the existing grant balance remains unchanged.

8. Notes

The use case supports the tracking of research grant expenses and the calculation of the remaining grant balance.